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HEALTH RISK ASSESSMENT (HRA) PROCEDURE

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1.0 PURPOSE

The purpose of the health risk assessment (HRA) procedure is to establish a systematic risk assessment methodology to:

1. Identify health hazards and potential exposure in operation.
2. Prevent and mitigate potential occupational health disease and illnesses.
3. Protect workers from adverse health effects of health hazards arising from the operation.
4. Review of existing control measure.
5. Comply with corporate standard, Qatar law and relevant occupational safety and health standard to protect workers health.
6. Enable decision making on improvement of existing control measure if required.

The HRA methodology is based on QatarEnergy HRA corporate standard and established Industrial Hygiene method and reference. The HRA Process is applicable to the life cycle of any operation.

2.0 SCOPE AND APPLICABILITY

The HRA will specifically only cover health hazards such as:

1. Chemical hazardous to health. Chemical that only hazardous by way of Physicochemical/safety hazard & environmental hazard is not included.
2. Physical hazard such as noise, vibration, heat, cold, ionizing and non-ionizing radiation.
3. Biological hazard such as legionella, bacteria, virus, mites, pathogens, and insects' bite/stings, etc.
4. Ergonomic Risk Factor/ Hazards- manual handling, office ergonomic etc.
5. Psychosocial Hazards.

The HRA are applicable for all operation in OI. Including routine and non-routine operation. This should be applicable to the life cycle of any operation. This includes:

Phase in Facility Lifecycle	Applicable Requirement
New activities and development.	To review detailed design stage to ensure HRA proactively conducted to minimize health risk in the new development. This include HRA of the detailed design & engineering, construction, commissioning and start up
Existing operations	Shall be conducted for routine and non – routine activities and development.

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Phase in Facility Lifecycle	Applicable Requirement
	Shall assess all health risk including those arising from short duration exposure to health hazards.
Change to existing activities	HRA shall be reviewed or revised when there are changes that effect the health hazard or risk of: I. Operation, II. Maintenance, III. Process, IV. Control V. Facilities, or VI. Hazard classification information. Management of Change (MOC) Process should be incorporated into the periodic review of HRA.

3.0 ROLES AND RESPONSIBILITIES

All employees have the responsibility to cooperate and participate in the HRA process.

3.1 OPERATION/ MAINTAINANCE/ PROJECT MANAGEMENT

- Responsible to ensure HRA is carried out for all activities under their management responsibilities. This includes routine and non-routine activities.
- Facilitate the implementation of HRA.
- Provide adequate resources to deliver HRA this includes participation of all personnel involve in HRA, providing information, supporting in the proposed corrective action.

3.2 INDUSTRIAL HYGIENIST

- To ensure qualitative HRA done every three years and coordination to perform follow up quantitative or specific HRA.
- To consolidate advice or further recommendation to control health risk based on HRA done.

3.3 HRA TEAM

- To conduct HRA as requirement or methodology outline by Industrial Hygienist.
- Equipped with strong knowledge and experience to conduct HRA.
- Must be competent to perform HRA required.
- Review the effectiveness of control measure for the HRA.

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- The HRA Team can either be led by internal Industrial Hygienist or appointed specialist health risk assessor appointed by QatarEnergy.

3.4 EMPLOYEE

- To participate and cooperate with HRA and proposed control measure.
- To cooperate during HRA by working hand in hand with HRA team and Industrial hygienist.
- To highlight any valid or potential health hazard concern or symptom that he/she felt arises from occupational health hazard exposure at work.

4.0 PROCEDURAL STEPS

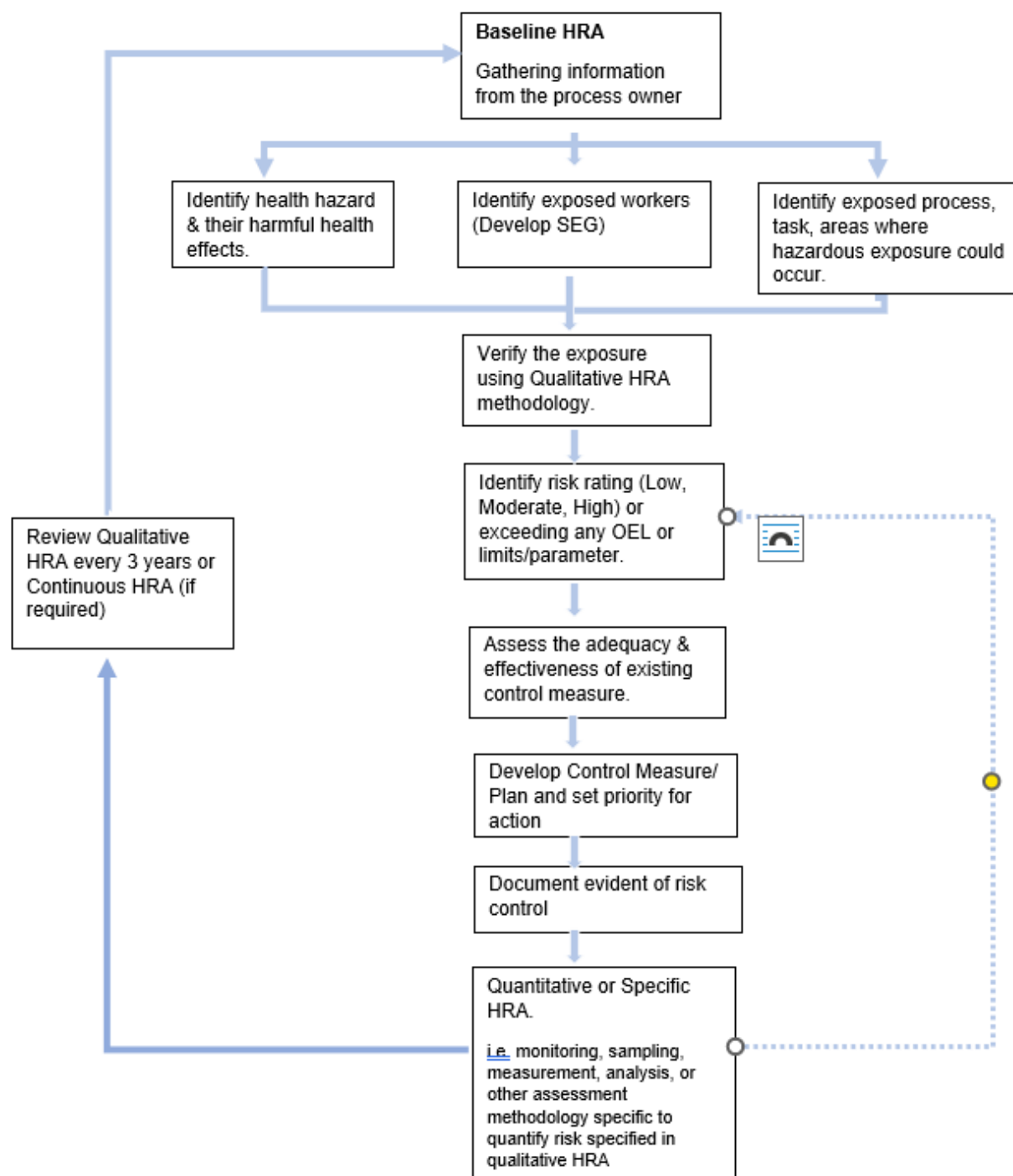
4.1 HEALTH RISK ASSESSMENT (HRA PROCESS)

HRA framework are proposed to ensure systematic risk assessment process for health hazard implemented. HRA need to be review every (3) three years. The outcome of the HRA would be a main qualitative HRA report and health risk summary to guide operation on health hazard and its control measure.

The HRA will cover every task and process in the workplace that exposes workers to health hazards. This includes routine and non-routine task.

The HRA process as follow.

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4.2 GATHERING INFORMATION

Information required for the HRA will include:

1. Site visit of the work area & observation of task (if possible)
2. Interview of relevant workers.
3. Reviewing document required.

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The list of documents required for HRA are as follow.

- a) List of chemicals being used in each workplace.
- b) Safety data sheet of chemicals.
- c) Process flowchart for workplace.
- d) Layout plan of work area.
- e) Description of task and processes for each SEG. This includes number of workers in the group, working hours, list of chemicals being handle, list of potential health hazard etc.
- f) Standard operating procedure relating to control of health hazards.
- g) Document related to Engineering control equipment such as Local Exhaust Ventilation (LEV), design parameters and record of inspections or maintenance.
- h) Personal Protective Equipment procedure or guidance.
- i) Personal exposure monitoring program data such as personal chemical exposure monitoring, heat stress measurement, radiation dose measurement or any quantitative data that can be used to measure exposure or the protection of health risk of the workplace (if any).
- j) Health surveillance & medical surveillance program and occupational disease or illness records. (if any).
- k) Other document that supports the protection of health of workers in the workplace such as: food safety inspection, potable water safety records, indoor air quality related maintenance record and program, etc.

This list is not exhaustive and additional document will be requested based on-site visit and interview done within the scope of HRA.

Similar Exposure Group (SEG) is defined workers of similar risk group. The characterization must be fulfilling two basic requirements.

- a. Work similarity – similar potential for health hazard exposure.
- b. Similarity with respect to exposed to similar health hazards. This could be chemical, physical hazard, biological hazard etc.

Example of SEG is operation technician, maintenance technician, rigger, kitchen staff, Living quarter personal etc.

SEG can be identify by:

- a. walk through inspection to identify person exposed,
- b. Acquiring each department/work area list of jobs, task and processes.
- c. Interview of the supervisor and workers regarding work practice.

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- d. Identification of health hazard exposure exposed, used, or released for the task carried out at the work areas.

Use the established HRA Form in the appendix to fill SEG description. Please refer to Industrial Hygienist for detail HRA Form to be used.

The outcome of HRA finding and control measure shall be incorporated into existing safe working system or procedure to ensure adequate control of health hazards.

HRA can only conducted by Industrial Hygienist or appointed personal with relevant competency or credential to assess occupational health risk exposure.

4.3 QUALITATIVE HRA

Qualitative HRA shall be done as an initial baseline health risk management process. Risk Matrix are developed based Qatar Energy corporate standard of health risk assessment and jointly develop by Industrial Hygienist based on the initial qualitative HRA.

The Qualitative HRA shall follow risk assessment matrix specified in this procedure and jointly develop by Industrial Hygienist or established methodology as a agreed by Industrial Hygienist.

Specific Risk Matrix should be referred to when conducting qualitative HRA.

Please refer to table below:

Qualitative Risk Matrix Based on Health Hazards	Refer to
Chemical Hazardous to Health	Appendix B – Risk Matrix for Chemical Hazardous to Health.
Physical, Biological, Ergonomic and Psychosocial Hazards	Appendix A – Risk Matrix Based on Health Effect, Probably and Scenario.

Other qualitative HRA methodology shall be consulted and approved by Industrial Hygienist.

4.4 QUANTITATIVE & SPECIFIC HEALTH RISK ASSESSMENT (HRA)

Based on the qualitative HRA, further quantitative or specialized HRA can be conducted based on the need further quantify the risk and specifying potential control measures.

The purpose is to further quantify the exposure and decide on appropriate control measure. Example for quantitative and specific HRA are:

No	Quantitative & Specific HRA	Purpose
1	Chemical exposure monitoring or air monitoring	Further quantify chemical exposure for individual or area

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No	Quantitative & Specific HRA	Purpose
2	Ergonomic Risk Assessment (Either specialized qualitative assessment or quantitative measure)	Further specify risk for ergonomic. This could be specialized for office or manual handling.
3	Noise Risk Assessment (include personal noise dosimeter & area measurement)	Quantification of personnel noise dose and identification of noise source in operation
4	Indoor Air Quality (IAQ) measurement	Measurement of IAQ parameter to ensure compliance.
5	Heat Stress (Heat Index) Measurement	Measurement of environmental heat stress or heat index in the summer season.
6	Food and water safety related laboratory analysis	To ensure the food and water safety aspects.
7	TLD Dosimeter, Personal Dosimeter and Radiation monitoring measurement	To ensure dose of radiation measured and survey possible radiation or NORM contamination.

The list above is not exhaustive and may be added based on the initial qualitative HRA and Industrial Hygienist advice, plan, and justification.

Quantitative exposure assessment involved with both personal and area monitoring; this may include various direct reading measurement to quantify exposure. Example, Hydrogen sulphide(H₂S) monitor, sound level meter, radiation survey meter, Direct reading gas tubes etc. This also include measurement of personnel dose exposure to chemicals or noise or radiation using various monitoring methodology.

Only appropriate and valid monitoring methodology (including analytical techniques) should be used and reference in HRA. All lab analytical should only be done by accredited laboratory with suitable and valid methodology.

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5.0 REFERENCES TO OTHER DOCUMENTS

TABLE 1: REFERENCES TO OTHER DOCUMENTS

Document Title	Document Number	Internal/External Document (if any)	Document Referencing
Corporate Standard for Health Risk Assessment	CORP- OHH-STD- 041	Internal	Upwards
Health Risk Assessment & Management	IOGP Report 384	External	N/A

6.0 DEFINITIONS

TABLE 2: KEY TERMS DEFINED

Term	Definition
OEL	Occupational Exposure Limit is maximum allowable concentration of a hazardous substance in a workplace. There are few type of OEL such TWA (time weighted average), STEL (short term exposure limit), MEL(maximum exposure limit) & ceiling limit (single/max value that should never be exceeded). There are other terminology being to represent OEL such as Permissible exposure limit (PEL), Threshold limit value (TLV).



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APPENDIX A – RISK MATRIX BASED ON HEALTH EFFECT RATING, PROBABILITY, LIKELIHOOD AND EXPOSURE.

	Rating level	A	B	C	D	E
	General	Very unlikely	Unlikely	Possible	Likely	Very Likely
	Historical	Never heard of in the industry operation	Has occurred in industry operation	Has occurred in the organization or similar operations	Happen about yearly in similar organizations worldwide	Happens about monthly in similar organization worldwide
	Descriptive	A freak combination of factors would be required for an incident to result	A rare combination of factors would be required for an incident to result	Could happen when additional factor is present otherwise unlikely to occur	Not certain to happen but an additional factor may result in accident	Almost inevitable that and accident would result
Potential Severity (Health effect rating)	Industrial Hygiene	<= 1% of OEL Exposures are negligible	1-10% of OEL Exposures are highly controlled	10-50% of OEL Exposures are well controlled	50-100% of OEL Exposure area not adequately controlled.	>= 100% of OEL Exposures are poorly controlled.
No health effect/injury/illness	0	Very Low Risk				
Slight health effect/injury/illness	1					
Minor injury or health effect	2		Low Risk			
Major injury or health effect	3			Medium Risk		
Single fatality/permanent total disability	4				High Risk	
Multiple fatalities	5					

Adopted from Qatar Energy Corporate Standard for Health Risk Assessment & IOGP Report 384

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Table 1; Description for Health Effect Rating(HER) Categories.

Class	HER / Severity	Description of Health effect	
Minor Incident	1 (Insignificant)	Slight Injury or Health Effect	• No treatment case of first aid case • occupational illness that result in noticeable discomfort, minor irritation or transient effect after exposure stops
	2 (Minor)	Minor Injury or Health Effect	• Medical treatment case • Lost workdays case or restricted work case, where the duration equals or less than 4 calendar days. • Occupational illness with reversible health effects such as food poisoning and acute dermatitis i.e. food poisoning, dermatitis, heat rash, heat cramp, heat exhaustion, irritant agents, defatting agents, any food poisoning bacteria, nonionizing radiation
Major Incident	3 (Moderate)	Major Injury or Health Effect	• Agents capable of irreversible health damage without loss of life. • Lost workdays case or restricted work case, where the duration more than 4 calendar days. • injury which results in permanent partial disability. • Occupational illness with irreversible health effects which result in permanent partial disability such as occupational noise induce hearing loss, chronic back disorder or repetitive strain injury i.e. Noise induced hearing loss (NIHL), musculoskeletal disorders, heat stroke, mental stress etc.

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Class	HER / Severity	Description of Health effect	
	4 (Major)	Single Fatality or Permanent Total Disability	● Fatality due to injury or occupational illness ● Injury which results in permanent total disability ● Occupational illness including cancer with irreversible health effect which result in permanent total disability such as asbestosis, silicosis, heart or kidney failure due to exposure i.e. heat stroke, carcinogen, cold stress, psychosocial hazard and ionizing radiation.
	5 (Catastrophe)	Multiple* Fatalities or Permanent Total Disability	● Fatalities due to injury or occupational illness ● Occupational illness including cancer with irreversible health effect which result in permanent total disability such as asbestosis, silicosis, heart or kidney failure due to exposure. * occurrence of multiple fatalities or permanent total disability caused by one single incident. * occurrence illness including cancer to effected works who exposed to similar hazard in a same operating unit or similar exposure group i.e. chemicals with acute toxic effects (hydrogen sulphide, carbon monoxide) or known human carcinogens (large, exposed population).

**HEALTH RISK ASSESSMENT (HRA) PROCEDURE****DOC NO: IS-HSE-PRC-017****REV. 00****TABLE 2: EXPOSURE EFFECT (LIKELIHOOD) RATING**

Exposure Rating	Exposure Band (OEL)	Definition	Likelihood
Very Low (1)	<= 1% of OEL (Occupational Exposure Limit)	Exposures are negligible	A (Very unlikely)
Low (2)	1-10% of OEL	Exposures are highly controlled – employees have minimal exposure	B (Unlikely)
Medium (3)	10-50% of OEL	Exposures are well controlled – employees have frequent contact at low concentrations and rare contact at high concentrations.	C (Possible)
High (4)	50-100% of OEL	Exposure area not adequately controlled - employees have frequent contact at low concentrations and infrequent contact at high concentrations.	D (Likely)
Very High (5)	>= 100% of OEL	Exposures are poorly controlled – employees often have contact at high or very high concentrations	E (Very likely)



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APPENDIX B – RISK MATRIX FOR CHEMICAL HAZARDOUS TO HEALTH.

		Exposure Rating (ER)				
		1	2	3	4	5
		<= 1% of OEL Exposures are negligible	1-10% of OEL Exposures are highly controlled	10-50% of OEL Exposures are well controlled	50-100% of OEL Exposure area not adequately controlled.	>= 100% of OEL Exposures are poorly controlled.
Hazard Rating (HR)	1	RR= 1	RR= 2	RR= 3	RR=4	RR=5
	2	RR= 2	RR=4	RR=6	RR=8	RR=10
	3	RR= 3	RR= 6	RR= 9	RR= 12	RR= 15
	4	RR= 4	RR= 8	RR= 12	RR= 16	RR=20
	5	RR= 5	RR= 10	RR= 15	RR= 20	RR= 25

Risk Rating (RR) is determined by **RR = HR (Hazard Rating) X ER (Exposure Rating)**.

Low risk	RR= 1 to RR = 4
Moderate risk	RR= 5 to RR = 12
High Risk	RR= 15 to RR = 25

Detail Methodology for Chemical Health Risk Assessment available separately with Industrial Hygienist.

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APPENDIX C – REVISION HISTORY LOG

TABLE 3: REVISION HISTORY LOG

Revision Number: 00**Date: 13/12/2023**

Item Revised	Revision Description	Page No.
Rev.00	Not applicable	Not applicable
Remarks: Initial release.		

APPENDIX D – LIST OF ABBREVIATIONS

TABLE 4: LIST OF ABBREVIATIONS

Abbreviation	Description
HRA	Health Risk Assessment